HH-P-00106d(NAVY-SHIPS)
18 May 1956

INTERIM FEDERAL SPECIFICATION

PACKING MATERIAL, FLAX OR HEMP

This interim Federal specification has been developed by the Department of the Navy, Bureau of Ships, based upon currently available technical information. It has not been approved for promulgation as a revision of Federal Specification HH-P-106c, dated 14 July 1948 and is subject to modification. This specification is in interim form only, and pending its promulgation by the administrator, General Services, as a regularly coordinated Federal specification, it may be used in procurement.

1. SCOPE

1.1 Scope. - This specification covers flax or hemp packing material for use in hot or cold water, brine and fuel oil systems (see 6.1).

1.1.1 Federal specification coverage. - This specification does not cover all types, classes and sizes of flax or hemp packing material, but only those most generally used by the Federal Government.

1.2 Classification. - Flax or hemp packing material shall be of the following classes, as specified (see 6.2):

Class 1 - Flax.
Class 2 - Hemp.
Class 3 - Flax and hemp.

1.2.1 Sizes. - The packing material shall be furnished in the sizes specified (see 6.2).

FED. SUP. CLASS
5330

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

O-E-751 - Ether, Petroleum; Technical-Grade.
NN-B-591 - Boxes; Fiberboard, Wood-Cleated for Domestic Shipment).
SS-G-659 - Graphite; Lubricating, Flake.
VV-L-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling, and Testing.
CCC-T-191 - Textile Test Methods.
LLL-B-631 - Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY

JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes; Wood Cleated; Solid Fiberboard.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-B-121 - Barrier Material, Greaseproofed, Flexible (Waterproofed).
MIL-B-10377 - Boxes; Wood-Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material. - The packing materials shall be composed of cleaned flax or hemp (cannabis sativa) fibers and a pure lubricant free from rancidity and suitable for the purpose intended.

3.2 Lubricant. - The packing material shall be thoroughly and evenly impregnated with the lubricant. The finished packing material shall contain not less than 35 nor more than 45 percent, by weight, of lubricant (see 4.3.2).

3..2..1 Oleic acid. - The lubricant shall contain not more than the equivalent of 5.5 percent oleic acid when tested as specified in 4.3.3.

3.2.2 Saponification number. - The saponification number of the lubricant, when tested as specified in 4.3.4, shall be not less than 185 nor more than 198.

3.3 Fiber. -

3.3.1 Type of fiber. -

3.3.1.1 Class 1, flax. - Class 1 packing material shall be made from flax fibers.

3.3.1.2 Class 2, hemp. - Class 2 packing material shall be made from hemp fibers.

3.3.1.3 Class 3, flax and hemp. - Class 3 packing material shall be made from a mixture of flax and hemp fibers.

3.3.2 Fiber strength. - The strength of the fiber shall be not less than 100 kilograms per gram per centimeter of length when tested as specified in 4.3.5.

3.4 Construction. - Packing material shall be either braided and squared, or rounded.

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3.5 Size and weight. - The packing shall be furnished in the sizes shown in table I, as specified (see 6.2). The weight per linear yard of square packing, after the lubricant has been extracted shall be not less than that shown in table I. The weight of round packing is found by multiplying the weight of the corresponding size of square packing by 0.8:

Table I - Size and weight of packing.

Size of packing	Weight per linear yard, minimum
Inches	Pounds
1/8	0.012
3/16	.030
1/4	.054
5/16	.082
3/8	.12
7/16	.16
1/2	.21
9/16	.27
5/8	.33
11/16	.40
3/4	.48
13/16	.57
7/8	.66
15/16	.75
1	.84
1- 1/8	1.08
1- 1/4	1.35
1- 1/2	1.80
1- 3/4	2.25
2	2.65

3.6 When packing coated with graphite is required (see 6.2), it shall be in accordance with Specification SS-G-659.

3.7 Workmanship. - The workmanship shall be first class in all respects.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.-

4.1.1 Inspection lot.- For purposes of sampling, inspection and tests, a lot shall consist of all finished packing material of one class and size, produced under essentially the same conditions, and offered for delivery at one time.

4.1.2 Sampling for visual and dimensional inspection.- A random sample of coils or spools shall be selected by the Government inspector from each inspection lot offered for Government inspection of visual and dimensional characteristics in accordance with Standard MIL-STD-105 at inspection level II as shown in table II.

Table II - Sampling for visual and dimensional inspection
AQL (approx.) = 2.5 percent defective.

Number of coils or spools in lot	Number of coils or spools in sample	Acceptance number (defectives)	Rejection number (defectives)
65 and under	7	0	1
66 to 180	25	1	2
181 to 300	35	2	3
301 to 500	50	3	4
501 to 800	75	4	5
801 to 1300	110	6	7
1301 and over	150	8	9

4.1.3 Sampling for lot acceptance tests.- The Government inspector shall select at random from each inspection lot four representative coils or spools of packing materials. Each sample coil or spool shall be selected from a different box. From each sample coil or spool, a 24-inch specimen shall be cut off for the tests specified in 4.3. For 1/8 and 3/16 inch sizes, the Government inspector shall select a sample 24 inches long of the untreated flax or hemp from the lot used in the manufacture of the packing material for the test specified in 4.3.1. Unless otherwise specified in the contract or order, these tests shall be conducted at the Naval Engineering Experiment Station.

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4.2 Inspection and tests. -

4.2.1 Visual and dimensional inspection. - Each of the sample coils or spools selected in accordance with 4.1.2 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any coils or spool in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective coils or spools in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection, provided the contractor has repaired or removed all nonconforming material. The Government inspector shall again examine samples from such resubmitted lots to verify compliance with this specification.

4.2.2 Lot acceptance tests. - Each sample specimen selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.3. If any specimen tested is found to be not in conformance with this specification, the lot which it represents shall be rejected. Rejected lots may be offered again for testing only after the contractor, after being informed of the reasons for rejection, has so reworked the entire lot as to remove all nonconforming material.

4.3 Tests. -

4.3.1 Weight. - Each specimen shall be carefully weighed and measured. From the weight and measurement obtained here and the lubricant content of the packing material obtained in 4.3.2, the average weight per linear yard of the packing material after the lubricant has been extracted shall be calculated.

4.3.2 Lubricant. - A specimen weighing approximately 25 grams shall be carefully unbraided into rovings, placed in a soxhlet extractor, and extracted with petroleum ether for 3 hours. The soxhlet shall fill and empty not less than three times per hour. The petroleum ether shall be evaporated from the extract, and the extract shall be dried for 1 hour at a temperature of 100° to 105°C., cooled and weighed. The amount of lubricant shall be calculated as follows:

$$\text{Lubricant, percent} = \frac{\text{weight of extract}}{\text{weight of specimen of packing}} \times 100$$

The petroleum ether used for extraction shall conform to Specification O-E-751.

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4.3.3 Oleic acid.- Accurately weigh 10 grams of the lubricant from 4.3.2 into a flask, add 50 milliliters of 95 percent alcohol which has been neutralized with weak caustic soda, and heat to the boiling point. Agitate the flask thoroughly in order to dissolve the free fatty acids as completely as possible. Titrate while hot with aqueous tenth-normal alkali, free from carbonate, using phenolphthalein, alkali blue, or turmeric as an indicator. Agitate thoroughly after each addition of alkali. To express results as percentage of oleic acid, use the following equation:

1 milliliter of tenth-normal alkali = 0.0282 gram of oleic acid.

Note.- In the above equation, alkali, 1 milliliter of which is equivalent to 0.5 percent of oleic acid, may be used.

4.3.4 Saponification number.- The saponification number of the lubricant shall be determined as specified in method 5401 of Specification VV-L-791.

4.3.5 Fiber strength test.- A specimen of the packing shall be carefully unbraided into rovings. The lubricant shall be thoroughly washed from the specimen with gasoline and the packing allowed to dry by evaporation. A length of 9 centimeters shall be cut from each of three rovings and each of these lengths shall be tested as follows: Weigh accurately on a chemical balance. Carefully untwist and straighten out the fibers, making them as nearly parallel as possible. Cement each end of the specimen with collodion for not more than 1/2 inch, care being taken not to twist the fibers during the cementing process. A fabric testing machine (Spec. CCC-T-191) operating at a speed of 12 ± 2 inches per minute, shall be used to determine the breaking strength. One-half inch of each end of the specimen shall be gripped in the clamps of the machine. The tensile strength of the fiber shall be calculated to kilograms per gram per centimeter of length. The formula used for the calculation shall be as follows, the average strength of the three specimens shall be reported as the tensile strength:

$$K = \frac{P \times L}{G}$$

Where:

K = tensile strength in kilograms per gram per centimeter of length.
P = breaking strength of specimen in kilograms.
L = length of specimen in centimeters.
G = weight of specimen in grams.

On the 1/8-inch and 3/16-inch sizes, this test will not be made on the finished packing but on samples of the flax or hemp from the lot used in its manufacture.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level C. - Flax or hemp packing shall be packaged in accordance with the manufacturers commercial practice.

5.1.2 Level A. -

5.1.2.1 Spools. - Unless otherwise specified in the contract or order, packing in diameters of 5/16 inch or less, shall be uniformly coiled on spools as follows:

Size of packing	Length of spool
Inch	Linear yards
1/8	200
3/16	150
1/4	134
5/16	120

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5.1.2.2 Coils. - Unless otherwise specified in the contract or order, packing in diameters of 3/8 inch or greater shall be uniformly coiled in individual lengths as follows:

Size of packing	Length of each coil	Number of coils per box
Inches	Linear yards	
3/8	9	8
7/16	8	8
1/2	7	6
9/16	6	6
5/8	5	5
11/16	5	5
3/4	5	4
13/16	5	4
7/8	4-1/2	3
15/16	4-1/2	3
1	4-1/2	3
1- 1/8	4	3
1- 1/4	4	2
1- 1/2	4	2
1- 3/4	4	1
2	4	1

5.1.2.3 Packaging. - Each set of coils or each spool shall be wrapped in greaseproof barrier material conforming to grade A of Specification MIL-B-121 and their unit packaged in folding boxes, set-up boxes or fiberboard boxes conforming to Specification PPP-B-566, PPP-B-676, LLL-B-631, LLL-B-636 or JAN-P-108, respectively.

5.2 Packing. -

5.2.1 Level C. - Flax or hemp packing material packaged in accordance with level A or C as specified (see 6.2) shall be packed in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

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5.2.2 Level B. - Packing material packaged in accordance with level A or C as specified (see 6.2) shall be packed in wood-cleated fiberboard, wood-cleated kraft veneer board, wood-cleated plywood or fiberboard boxes conforming to Specification NN-B-591, MIL-B-10377 (domestic type), PPP-B-601 (domestic type), or the special requirements of Specification LLL-B-631 or LLL-B-636. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds.

5.2.3 Level A. - Packing material packaged in accordance with level A or C as specified (see 6.2) shall be packed in wood-cleated fiberboard, wood-cleated plywood, wood-cleated kraft veneer board, or fiberboard boxes conforming to Specification JAN-P-103, PPP-B-601 (overseas type), MIL-B-10377 (overseas type), or JAN-P-108. Boxes shall be closed and strapped in accordance with the applicable box specification and appendix thereto. Wood or wood-cleated boxes shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds, of fiberboard boxes 70 pounds.

5.3 Marking. -

5.3.1 Non-Military marking. -

5.3.1.1 Packages. - Unless otherwise specified in the contract or order, packages shall be marked with the name of the item, the class and size, the quantity contained therein, and the name of the manufacturer.

5.3.1.2 Shipping containers. - Unless otherwise specified in the contract or order, shipping containers shall be marked with the name of the item, the class and size, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

5.3.2 Military marking. - In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Packing material covered by this specification is for use on rods and valve stems under the following conditions:

(a) Hot or cold water, or brine up to 300 pounds pressure.
(b) Fuel oil up to 500 pounds pressure.

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Paper material not as specified

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Size required (see 1.2.1 and 3.5).
(d) If coated packing is required (see 3.6).
(e) Whether packaging should be in accordance with level A or C or packing in accordance with level A, B or C (see 5.1 and 5.2).

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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